

Introduction to Classification & 1R

Data Mining
Howles

Classification

- Given a set of preclassified examples, build a model or classifier to classify new cases
- Classification is a form a supervised learning because the classes are already known

Examples

- Is this customer a good credit risk?
- Is this credit card transaction suspicious?
- Given a set of symptoms, does the patient have the flu?

Examples

- Should we send this customer the new marketing brochure (is he/she likely to purchase the product)?
- Is the image in the sky an airplane, helicopter or missile?
- Face recognition. Uses facial measurements (distance from nose to mouth, shape of head, ...) to identify people

Classification

- Classification must result in classifications that are mutually exhaustive and exclusive
 - Exhaustive – all must be assigned
 - Mutually exclusive – assigned to exactly one class

Classification Algorithms

- Classifiers can be rule-based, tree-based, neural nets, ...
- We will look at rule and tree-based methods

William Ockham

- 14th Century Monk
- His guideline states that simpler methods yield the best results
 - When possible, use the fewest assumptions, fewest entities that will sufficiently answer your question
- His guideline: Ockham's Razor (sometimes spelled Occum or Occam)

OR

- (Zero R)
- Mentioned here for completeness
- Take the majority rule
- Very simplistic, usually too simplistic for practical use

1R

- 1R – “1-Rule”
- Select a single attribute and pick the one that works the best (fewest errors)
- Simple, and often times the simple approaches work the best
- Actual success depends on the domain

1R

- Inference rules
- Easy; fast and quite accurate
 - 1R's simple rules perform no worse than other more complex decision trees
- 1 Rule for every attribute value
- Essentially learns a 1-level decision tree

1R

- One branch for each value
- Each branch assigns the most frequent class
- Error rate: Proportion of instances that don't belong to the majority class of their corresponding branch
- Select attribute with lowest error rate

1R Pseudocode

For each attribute A

 Make a rule for each value of A by:

 Counting the class appearances

 Finding the most frequent class

 Making the rule assigned to that class the
 attribute value A

 Calculate the error rate

Select the rules with the smallest error rate

The Weather Database

- 1R using the weather database

Outlook	Temp	Humidity	Windy	Play
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

Outlook

- Look at the outlook attribute
- How many of each:
 - Sunny -> play/no play
 - Rainy -> play/no play
 - Overcast -> play/no play

Outlook

- How many of each:
 - Sunny -> play/no play 2/3
 - Rainy -> play/no play 3/2
 - Overcast -> play/no play 4/0

Outlook

- Now look at most occurring value and calculate errors:
 - Sunny -> play/no play 2/3 - Pick NO PLAY and we will be wrong 2 times
 - Rainy -> play/no play 3/2 – Pick PLAY and we will be wrong 2 times
 - Overcast -> play/no play 4/0 – pick Play and we will be wrong 0 times
- For Outlook, the total errors is 4

Temp

- How many of each:
 - Hot -> play/no play
 - Mild-> play/no play
 - Cool-> play/no play

Temp

- How many of each:
 - Hot -> play/no play 2/2
 - Mild -> play/no play 4/2
 - Cool -> play/no play 3/1

Temp

- Now look at most occurring value and calculate errors:
 - Hot $\rightarrow$ play/no play 2/2 – Pick either one (arbitrary pick when a tie) and we will be wrong 2 times
 - Mild $\rightarrow$ play/no play 4/2 – Pick PLAY and we will be wrong 2 times
 - Cool $\rightarrow$ play/no play 3/1 – Pick PLAY and we will be wrong 1 time
- For Temp, the total errors is 5

Humidity and Windy

- You derive the rules

Attribute	Rules (Most frequent)	Errors	Total Errors
Outlook	Sunny -> No (Sunny yes = 2; no = 3; 3 is the most so we use NO and are wrong 2/5)	2/5	4/14
	Overcast -> Yes	0/4	
	Rainy -> Yes	2/5	
Temp	Hot -> No (Tie with Hot -> Yes so randomly pick)	2/4	5/14
	Mild -> Yes	2/6	
	Cool -> Yes	1/4	
Humidity	High -> No	3/7	4/14
	Normal -> Yes	1/7	
Windy	False -> Yes	2/8	5/14
	True -> No	3/6	

Note a few ties Temp:Hot->No and Windy:True-> No

The Result

- Looking at the results, we would pick the rule with the smallest error rate
- In this case, either the Outlook or Humidity attributes

1R and Missing Data

- If missing data, 1R treats as another value and calculates the error
- Outlook: Overcast, sunny, rainy, “missing”

Problems with 1R

- Overfitting may result if we have a large number of categories/values – the rules may be too specific and may not fit future states
- We calculate a 0 error rate for unique fields – it overfits the data
- Ties are arbitrarily selected
- ***It is too simplistic for use in your projects; use only as a benchmark***

Performance Measurement

- Evaluate the accuracy of your classifier
- The correct answer may depend on the user and specific application
- Measurement is correctness of the algorithm or classifier selected

Performance Measurement

- Success may be defined as:
 - True positive - predicted to be in this class and is actually in it
 - False positive – predicted to be in this class but is not actually in it
 - True negative – predicted not to be in this class and is not actually in it
 - False negative – predicted not to be in this class but is actually in it

Class Predictions

- A matrix can be constructed showing the true/false positives and true/false negatives

True Positives	False Negatives
False Positives	True Negatives

Confusion Matrix

- If we have m possible classifications, create an $m \times m$ table showing the errors (estimated classification $\neq$ actual classification)
- Best solutions only have zero values outside the diagonal

Confusion Matrix

Example

	a	b	c	←- classified as
a	5	0	0	
b	4	0	0	
c	3	0	12	

We are trying to assign to 3 classifications.

'a' is good – all classified correctly

'b' is not so good; b classified as a

'c' is so-so 12 correct; 3 classified as a

Confusion Matrix

- Construct a confusion matrix to evaluate the correctness of your algorithm
- Most tools will build the matrix for you